POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2024
Programme: BE | Full Marks : 100
Course: Applied Operating Systems (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain Operating System. List out different types of operating system. Explain any three of them in brief. 7

   b) Why process is heavyweight in comparison with thread? How Process Control Block (PCB) is used in process management and to handle context switching between processes? 8

2. a) Draw a Gantt Chart and find average turnaround time and waiting time of the following process applying SJF(Preemptive), SRTF, FCFS and Round Robin (quantum=3) scheduling algorithm. 8

Process	A	B	C	D	E
Arrival Time	0	4	4	3	2
Burst Time	4	6	5	8	7

   b) Explain the concept of semaphores as synchronization primitives. Explain how mutual exclusion and synchronization ensure correct execution in concurrent environments. 7

OR

What is race conditions and how they can lead to inconsistent or incorrect results in shared data access? Explain the basic requirement for the solution of critical section problem.

3. a) Define the critical section. The snapshot of the state is given below. Is this a safe state? If yes, what is its safe sequence? 8

Process	Allocation				Max				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P0	3	0	1	4	5	1	1	7	1	0	0	2
P1	2	2	1	0	3	2	1	1				
P2	3	1	2	1	3	3	2	1				
P3	0	5	1	0	4	6	1	2				
P4	4	2	1	2	6	3	2	5				

   b) Differentiate sequential file access method and random file access method. Explain different file allocation strategies. 7

4. a) What is page fault? Given below is the reference made to the following pages by a program: 1,3,1,5,7,7,3,7,4,9,8,1,6,3,4,2,5,8. Show the successive page residing in the four frames using replacement policy below: 8
   i. FIFO
   ii. LRU
   iii. Optimal

   Also calculate page fault.

   b) What is virtual memory? Explain the structure of page table with suitable diagram. 7

OR

Under what circumstances do page faults occur? Describe the actions taken by the operating system when a page fault occurs.

5. a) The disk track requests are: 125, 250, 298, 123, 13, 350 and 224. Assume that the last request is at track 150 and the head is moving towards track 0. Find out the total seek time for each of the disk scheduling algorithms below: 8
   i. SSF
   ii. C-SCAN

iii. FIFO

iv. SCAN

b) Define device controllers. Explain I/O software layers. 7

6. a) Discuss the design principles IPC mechanisms and security management of any operating system (LINUX, WINDOWS). 7

b) Explain the concept of the memory wall and discuss the major bottlenecks faced by operating systems in memory management. 8

7. Write short notes on: **(Any two)** 2×5

a) OS as a virtual machine

b) Goals of protection

c) LINUX Vs Windows OS

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2024

Programme: BE | Full Marks: 100

Course: Applied Operating System (New) | Pass Marks: 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Explain Operating System. List out different types of operating system. Explain any three of them in brief. 8

b) What is a process? Illustrate and define the different states of process with a neat diagram. 7

2. a) Define Critical Section and Race Condition. How Dinning Philosopher problem can be solved using semaphore? 7

OR

What are the usage of sleep and wakeup operations in process synchronization? Explain two-task solution to achieve mutual exclusion.

b) Explain Resource Allocation Graph. Consider a system with three processes P1 through P3 and four allocable resources. The total four resources types in the amount as E=(4,2,3,1). Suppose at time t0 following snapshot of the system has been taken: 8

i. What is the content of the need matrix?

ii. Is the system in safe state?

Process	Allocation Matrix				Process	Max Matrix			
	R1	R2	R3	R4		R1	R2	R3	R4
P1	0	0	1	0	P1	2	0	1	1
P2	2	0	0	1	P2	1	0	0	0
P3	0	1	2	0	P3	2	1	0	0

3. a) What is page fault? Given below is the reference made to the following pages by a program: 1,3,1,5,7,7,3,7,4,9,8,1,6,3,4,2,5,8. Show the successive page residing in the four frames using replacement policy below: 8

i. Second Chance

ii. LRU

iii. Optimal

b) Define file system. Explain linked list and i-node file system implementation. 7

4. a) Differentiate sequential file access method and random file access method. Explain different file allocation strategies. 7

b) Explain Direct Memory Access mechanisms. Explain the situations where DMA and programmed IO performed better than each other. 8

OR

Define device drivers. Explain three different ways of performing input and output.

5. a) The disk track requests are: 125, 250, 298, 123, 13, 350 and 224. Assume that the last request is at track 150 and the head is moving towards track 0. Find out the total seek time for each of the disk scheduling algorithms below: 8

   i. SSF
   ii. C-SCAN
   iii. FIFO
   iv. LOOK

b) What are the different components used in Windows Operating System. 7

6. a) Given the following set of information, calculate the average waiting time and average turn-around time using SJF(preemptive), RR (Quantum = 2) and priority(smaller number as higher priority) scheduling. Also describe which one is the best algorithm and why? 8

Process	Burst Time	Arrival Time	Priority
P1	8	0	3
P2	1	1	1
P3	3	2	2
P4	2	3	3
P5	6	4	4

b) What are the characteristic features of distributed operating system? Explain cloud operating system. 7

7. Write short notes on: (Any two) 2×5

   a) Virtual Machine
   b) Swapping
   c) RAID

POKHARA UNIVERSITY

Level: Bachelor
Semester: Spring
Year : 2025
Programme: BE
Full Marks : 100
Course: Applied Operating Systems (New)
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define operating system. Explain the structure of an operating system and the role of kernel data structures with examples. 8

b) Explain PCB and describe different fields of PCB and its role in context switching. 7

2. a) Given the following information's, draw GANTT charts and find the average waiting time and average turn-around time using STRF, RR (quantum=2), HRRN, FCFS. 8

Process	Arrival Time	Burst Time
A	0	9
B	4	6
C	5	7
D	7	12

b) Differentiate User Level and Kernel Level threads. Explain the multithreading models with appropriate diagrams. 7

3. a) Suppose we have two resources, A and B. A has 7 instances and B has 4 instances. Can the system the following processes without deadlock occurring? 8

Process	Allocate		Maximum need	
	A	B	A	B
P1	2	2	3	3
P2	1	0	4	2
P3	1	0	3	2
P4	0	1	1	1
P5	2	1	6	3

POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2025
Programme: BE | Full Marks : 100
Course: Applied Operating Systems (New) | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Elaborate on how the concept of a Virtual Machine (VM) is useful within the field of operating systems. Explain how multiprocessing OS is different from multiprogramming OS. 8

   b) Explain the lifecycle of a process with the help of a process state diagram. 7

2. a) Explain Banker's Algorithm for Deadlock Detection with an example. 8

   OR

   Explain Resource Allocation Graph. Consider a system with three processes P1 through P3 and four allocable resources. The total four resources types in the amount as E = (4, 2, 3, 1). Suppose at time t_0 following snapshot of the system has been taken:

   i. What is the content of the need matrix?
   ii. Is the system in safe state?

Process	Allocation Matrix				Process	Max Matrix			
	R1	R2	R3	R4		R1	R2	R3	R4
P1	0	0	1	0	P1	2	0	1	1
P2	2	0	0	1	P2	1	0	0	0
P3	0	1	2	0	P3	2	1	0	0

   b) Draw a Gantt Chart and find average turnaround time and waiting time of the following process applying SJF(Preemptive), SRTF and FCFS scheduling algorithm. Which algorithm is better and why? 7

Process	P1	P2	P3	P4	P5
Arrival Time	0	1	2	3	4
Burst Time	5	3	8	6	4

3. a) Differentiate binary semaphore with counting semaphore. Why are sleep and wakeup operations important in process synchronization? 8

   b) Define the critical section. Explain about the Dining Philosophers Problem along with its solution using appropriate synchronization mechanisms. 7

   OR

   Describe Peterson's algorithm and explain how it ensures mutual exclusion.

4. a) Explain Segmentation with Paging and illustrate it with a diagram. 7

   OR

   Define Demand Paging. Explain the different structures of the page table.

   b) Define file system. Explain contiguous, linked and i-node file allocation methods. 8

5. a) What is page fault? Consider the following page reference strings: 8

   7,0,1,2,0,3,0,4,2,4,0,3,2

   How many page faults would occur for each of the following page replacement algorithms:

   i. Second Chance
   ii. LRU
   iii. Optimal

   b) Differentiate between internal and external fragmentation? Explain TLB in detail. 7

6. a) The disk track requests are: 143, 89, 176, 130, 10, 150 and 190. Assume that the last request is at track 100 and the head is moving towards track 0. Find out the total seek time for each of the disk scheduling algorithms below: 8

   i. SSF
   ii. LOOK
   iii. FIFO

   b) Compare Linux and Windows operating systems in terms of kernel design, security, and system components. 7

7. Write short notes on: **(Any two)** 2×5

   a) RAID
   b) Cloud Operating System
   c) Principles of I/O Hardware

Explain how they help manage process execution and ensure that system resources are used efficiently.

b) Define Demand Paging. Explain the concept of paging with a suitable example. 7

OR

Define Page Fault. Why does excessive page faults (thrashing) occur? Describe the sequence of steps the OS takes when a page fault occurs.

4. a) Given below is the reference made to the following pages by a program: 2, 3, 2, 1, 5, 2, 4, 5, 3, 2, 5, 2, 1, 4, 6, 2, 3. Show the successive page residing in the three frames using replacement policy below. Also find page fault. 8
   i. FIFO
   ii. LRU
   iii. Optimal

   b) What are points to be consider in file system design? Explain different file allocation methods. 7

5. a) Explain TLB briefly with necessary diagrams. Differentiate between the concept of logical memory and physical memory. 7

   b) Suppose the following request queue (in order): 30, 145, 90, 10, 180, 75, 150, 5, 160, 85 with the head initially at track 95. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms? 8
   i. FCFS
   ii. SSTF
   iii. SCAN (initially moving upward)
   iv. C-SCAN (initially moving downward)

6. a) Discuss the design principles, ICP mechanism and security features in a modern OS of your choice. 8

   b) What are the major security issues faced by modern operating systems? Explain the methods used for their secure deployment. 7

7. Write short notes on: **(Any two)** 2×5
   a) System Call
   b) Interrupt driven I/O
   c) Security in Linux

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2015
Programme: BE Full Marks: 100
Course: Applied Operating System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define System call. Define the properties of following operating sytem. 8
   a. Real time
   b. Time sharing
   c. Parallel system.

   b) Why inter process communication is required? Explain. Differentiate between user and kernel thread. Draw figures to illustrate. 7
2. a) Define race conditions. What are the techniques for avoiding race condition? Why disabling interrupt method is unattractive to achieve race condition? Explain. 7

   b) From the following set of information, Determine the average waiting time and average turnaround time using FCFS, RR (Quantum = 5) and HRRN. 8

Process	Arrival Time	Service Time (Burst Time)
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

3. a) What is dispatcher and dispatch latency? Differentiate short-term scheduler, medium term scheduler and long-term scheduler. 7

   b) Consider the following Snapshot of a system: 8

Processes	Allocation ABCD	Max ABCD	Available ABCD
P0	0012	0012	1520
P1	1000	1750	
P2	1354	2356	
P3	0632	0652	
P4	0014	0656	

Answer the following questions using the Banker's algorithm:
   i. What is the content of the matrix need?
   ii. Is the system in a safe state? Also find the safe sequence.
   iii. If the request from process P1 arrives for (0, 4, 2, 0), can the request be granted immediately?

4. a) Define segmentation. Given memory partitions of 10k, 4k, 15k, 17k and 15k in order. How would each of first- fit, best fit and worst-fit algorithms place processed of 12k, 13k and 5k, in order? Which algorithm makes the best use of memory? 7

   b) What is page fault? Consider the following page reference strings: 7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1. How many page faults would occur for each of the following page replacement algorithms assuming 3 and 4 pages a frame? In each case which algorithm perform better? 8
   i. LRU page replacement
   ii. FIFO page replacement
   iii. Optimal page replacement
5. a) Disk requests come in to the disk driver for cylinders 10, 22, 44, 20, 2, 40, 6, and 38, in that order. A seek takes 2.1 msec per cylinder moved. How much seek time is needed for 8
   i. FCFS
   ii. SSTF
   iii. SCAN
   iv. CSCAN

   In all cases, the arm is initially at cylinder 20.

   b) Describe different memory allocation techniques in memory management. Explain Paging and TLB. 7
6. a) Describe about stable storage implementation and tertiary storage structure. 7

   b) Differentiate sequential file access method and random file access method. Explain different file allocation strategies. 8
7. Write short notes on: **(Any two)** 2×5

   a) Producer Consumer Problem

   b) Access Control List

   c) RAID

POKHARA UNIVERSITY

Level: Bachelor　　Semester: Spring　　Year : 2015
Programme: BE　　Full Marks: 100
Course: Applied Operating System　　Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) How System call is different from API? Describe various services provided by an operating system. Describe the actions taken by a Kernel to switch context between processes? 8
   b) Explain layered operating system. How process differs from thread? Explain. 7
2. a) Why inter-process communication and threading is required? Explain. 7
   b) Describe Peterson's solution. Define and explain the proposes of spinlock, mutex, semaphore, conditional variable and monitor. 8
3. a) Differentiate preemptive and non-preemptive scheduling algorithm? Consider the following set of processes, with the length of the CPU burst given in milliseconds. The processes are assumed to have arrived in the orde. P1, P2, P3, P4, P5, all at time 0. 8

Process	Burst Time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

   i. Draw Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non preemptive priority, and RR (quantum= 1).
   ii. What is the waiting time of each process for each of these scheduling algorithms?
   iii. What is the turnaround time of each process for each of the scheduling algorithms in part 1?

   iv. Which of the algorithms results in the minimum average waiting time (over all processes)?

   b) What are the conditions that may cause deadlock to arise? How does it differ from starvation? Explain with suitable example. 7
4. a) Define fragmentation. Given memory partitions of 10k, 4k, 15k, 17k and 15k in order. How would each of first- fit, best fit and worst-fit algorithms place processed of 12k, 13k and 5k, in order? Which algorithm makes the best use of memory? 7
   b) What is page fault? Consider the following page reference strings: 3, 3, 5, 4, 7, 1, 5, 5, 1, 4, 3, 7, 6, 3, 4, 1. How many page faults would occur for each of the following page replacement algorithms assuming 3 and 4 pages a frame? In each case which algorithm perform better? 8
      i. LRU page replacement
      ii. FIFO page replacement
      iii. Optimal page replacement
5. a) What is the role of device drivers? Write the principle of I/O System. 7
   b) Suppose that disk drive has cylinder numbered from 0 to 199. The head is at cylinder 91. The queue for services is as 99, 153, 47, 143, 167 and 67. What is the total head movement for each algorithms? 8
      i. FCFS
      ii. SSTF
      iii. SCAN
      iv. C-SCAN
6. a) Differentiate sequential file access method and random file access method. Explain different file allocation strategies. 8
   b) What do you mean by access methods? Describe directory structures with help of diagram. 7
7. Write short notes on: (Any two) 2×5
   a) TLB
   b) Blocking and non-blocking I/O
   c) Free space management in file system.

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2016
Programme: BE Full Marks: 100
Course: Applied Operating System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define operating system. List services provided by an operating system. 7

   b) Why creating a thread is cheaper than creating a process? Explain life cycle of the process with suitable diagram. 8

2. a) Why inter process communication is required? Explain semaphore based process synchronization method. 7

   b) Given the following information, draw the GANTT Charts for processor scheduling for HRRN, Preemptive Shortest Job First (SJF) and RR (Quantum = 2). Also, find the average waiting time, average turnaround time and average response time for all the cases. 8

Process	Arrival Time	Burst Time
P1	0.0	7
P2	3.0	4
P3	5.0	2
P4	6.0	4

3. a) Consider the following snapshots of the system having 10 instances of resource A, 5 instances of resource B and 7 instances of resource C. 8

Processes	Allocation A B C	Max A B C	Available A B C
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

Answer the following questions using the Banker's algorithm:

   i. Is the system in a safe state? Also find the safe sequence.

   ii. If the request from process P1 arrives for (1, 0, 2), can the request be granted immediately?

   b) What is paging? Consider the following page reference string: 7
   1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. How many page faults would occur for the following replacement algorithms, assuming three frames? In each case calculate fault ratio.

   i. LRU page replacement

   ii. FIFO page replacement

   iii. Optimal page replacement

4. a) Differentiate logical address with physical address. Explain coalescing and compaction. How Paging hardware with TLB does improves the performance during memory mapping. 8

   b) What is the role of device drivers? Write the principle of I/O System. 7

5. a) What is transfer time, seek time and rotational latency? Disk requests come in to the disk driver for cylinders 10, 22, 20, 2, 40, 6, and 38, in that order. A seek takes 6.25 msec per cylinder moved. How much seek time is needed for 8

   i. FCFS

   ii. SSTF

   iii. SCAN

   iv. CSCAN

   In all cases, the arm is initially at cylinder 20.

   b) What is interleaving and why it is required? Explain the types of interleaving. Describe the sector slipping mechanism of bad sector recovery in disk management. 7

6. a) Define file. Write about I/O request handling. Write about linked list file allocation method. 8

   b) Describe different file sharing semantics. Explain the strategies of free space management. 7

7. Write short notes on: **(Any two)** 2×5

   a) Real-Time Systems

   b) Producer Consumer Problem

   c) Segmentation

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2016
Programme: BE | | Full Marks: 100
Course: Applied Operating System | | Pass Marks: 45
| | Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is Operating System? Define it with regards to the Personal Computer System. How is it different from batch systems? 7
   b) What are kernel and user threads? What are the different thread states? Briefly explain the multithreading models. 8
2. a) What are deadlocks? What are the pre-requisites for the system to encounter deadlock? Explain in brief the various deadlock prevention mechanisms. 8
   b) When multiple processes need to cooperate, there is a choice between shared memory and inter-process communication (IPC). Compare and contrast these two techniques. What is the role of the operating system in each? 7
3. a) How does race condition occur in a process? Explain dining philosopher's problem as a classic synchronization problem. 7
   b) Consider following set of processes along with their burst time, arrival time and priorities. Calculate average waiting time and average turnaround time using following scheduling algorithms. 8
      i. FCFS
      ii. SJF
      iii. Priority (Preemptive)
      iv. HRRN

Process	Arrival Time	Burst Time	Priority
A	0	4	5
B	2	3	4
C	5	7	1
D	7	5	3
E	4	1	2
F	3	4	1

4. a) Explain different types of page replacement algorithms? Find out how many page faults occur in the following sequence of the reference string using FIFO, OPR, LRU page replacement algorithms using 3 frames. 7,0,3,5,1,3,7,5,0,3,7,1,3,1,7,0,5,3,1,7. 8
   b) Differentiate virtual page and a page frame. What is the difference between LRU and NRU page replacement algorithms? Explain. 7
5. a) Disk request come to the disk driver for cylinder 16, 18, 12, 6, 25, 38, 7 and 36 in that order. A seek take 2 micro sec per cylinder move. How much seek time is needed for 8
      i. FCFS
      ii. Closest Cylinder Next
      iii. C-Scan (Initially moving upward)
      iv. Scan (Initially moving downward)

   In all cases, the arm initially at cylinder 18. Also describe which one is best algorithm and why?
   b) Write down different file access methods. What are file allocation methods? Explain briefly. 7
6. a) How does OS handle the bad sectors in the disk? Explain. 5
   b) Describe how Paging leads to internal fragmentation. 5
   c) What is meant by sector sparing or forwarding? 5
7. Write short notes on: (Any two) 2×5
   a) System Calls
   b) Thread Scheduling
   c) File protection methods

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2017
Programme: BE Full Marks: 100
Course: Applied Operating System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define time sharing system. Describe various services provided by an operating system. Describe the actions taken by a Kernel to switch context between processes? 7

   b) How is a thread different from a process? Describe in detail about the different state transition model of a process. 8

2. a) Let five processes; A, B, C, D and E be on queue in a scheduler. They arrived in the queue at the instance; 0ms, 3ms, 3ms, 6ms, and 9ms respectively. The time they require to complete is 2 ms, 5ms, 10ms, 4ms, and 4ms respectively. Using FCFS, STRF and RR (quantum=2ms), evaluate the given scenario. On the basis of average wait time, average response time and average turnaround time, which algorithm is suitable? Describe. 8

   b) What is a race condition? Give an example of a race condition that could possibly occur when buying airplane tickets for two people to go on a trip together. 7

3. a) Consider the deadlock situation that could occur in the dining-philosophers' problem when the philosophers obtain the chopsticks one at a time. Discuss how the four necessary conditions for deadlock indeed hold in this setting. Discuss how deadlocks could be avoided by eliminating any one of the four conditions. 7

   b) Explain the different File Access Methods? Describe the different issues associated with contiguous file allocation. 8

4. a) What are page replacement algorithms? Find out how many page faults occur in the following sequence of the reference string using FIFO, OPR, MRU page replacement algorithms using 4 frames. 7,0,3,6,8,1,2,3,7,5,0,2, 3,7,1,3,1,4, 6, 4, 7,0,5,3,1,7. 8

   b) What is preemptive and non-preemptive scheduling algorithm? Explain Multi level queue and multi-level feedback queue scheduling with appropriate example. 7

5. a) Disk request come to the disk driver for cylinder 26, 18, 10, 66, 92, 38, 74 and 31 in that order. A seek take 2 micro sec per cylinder move. How much seek time is needed using: (Assume no. of cylinders: 100) 8
      i. FCFS
      ii. Closest Cylinder Next (SSTF)
      iii. C-Scan (Initially moving upward)
      iv. Look (Initially moving downward)

   b) What are tertiary storage devices? Explain about any two such devices. 7

6. a) Describe Coalescing and compaction. Given six memory partitions of 300 KB, 600 KB, 350 KB, 200 KB, 750 KB, and 125 KB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of size 115 KB, 500 KB, 358 KB, 200 KB, and 375 KB (in order)? Rank the algorithms in terms of how efficiently they use memory. 8

   b) Explain directory structure and its types. Explain about access control list (ACL). 7

7. Write short notes on: **(Any two)** 2×5
   a) System call
   b) DMA and Polling
   c) Demand Paging

POKHARA UNIVERSITY

Level: Bachelor	Semester: Spring	Year : 2017
Programme: BE		Full Marks: 100
Course: Applied Operating System		Pass Marks: 45
		Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are batch, time-sharing, parallel and real-time operating systems? Discuss their characteristics. 7

   b) What is Process? Draw and describe process state diagram. What types of operations are performed in a process? 8

2. a) What are the requirements for the solution of critical section problem? Explain the software solution for critical section problem. 7

   b) From the following set of information, determine the average waiting time and average turn around time using FCFS, RR (Quantum = 2) and HRRN 8

Process	Arrival Time	Service Time (Burst Time)
A	0	3
B	2	6
C	4	4
D	6	5
E	8	2

3. a) What is dispatcher and dispatch latency? Differentiate short-term scheduler, medium term scheduler and long-term scheduler. 7

   b) Consider the following Snapshot of a system: 8

Processes	Allocation ABCD	Max ABCD	Available ABCD
P0	0012	0012	1520
P1	1000	1750	
P2	1354	2356	
P3	0632	0652	
P4	0014	0656	

Answer the following questions using the Banker's algorithm:

i. What is the content of the matrix need?

ii. Is the system in a safe state? Also find the safe sequence.

iii. If the request from process P1 arrives for (0, 4, 2, 0), can the request be granted immediately?

4. a) Describe the actions taken by a kernel to context-switch between processes. Provide two programming examples in which multithreading doesnot provide better performance than a single-threaded solution? 7

   b) What is page fault? Consider the following page reference strings: 1, 3, 5, 3, 7, 1, 5, 3, 1, 2, 3, 7, 6, 3, 4, 1, 8. How many page faults would occur for each of the following page replacement algorithms assuming 3 and 4 pages a frame? In each case which algorithm perform better? 8

   i. LRU page replacement
   ii. FIFO page replacement
   iii. Optimal page replacement

5. a) Disk requests come in to the disk driver for cylinders 10, 22, 20, 2, 40, 6, and 38, in that order. A seek takes 6.25msec per cylinder moved. How much seek time is needed for 7

   i. FCFS
   ii. SSTF
   iii. SCAN(initially moving downward)
   iv. CSCAN(initially moving upward)

   In all cases, the arm is initially at cylinder 20.

   b) What are the advantages and disadvantages of supporting memory mapped I/O to device control registers? 8

6. a) What is bad sector? Describe the different bad block recovery mechanisms in disk management. 7

   b) How free-space management is done in file systems? Explain. 8

7. Write short notes on: **(Any two)** 2×5

   a) DMA
   b) Sequential File Access Method
   c) RAID

POKHARA UNIVERSITY

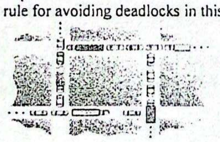
Level: Bachelor Semester: Fall Year : 2018
Programme: BE Full Marks: 100
Course: Applied Operating System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is the main advantage of the layered approach to system design? What are distributed, interactive, and parallel operating systems? Discuss their characteristics. 8
   b) What is the fundamental difference between a process and a thread? What is the biggest advantage and disadvantage of implementing threads in user space? 7
2. a) What is mutual exclusion? Show how mutual exclusion can be achieved using Peterson's Solution. 7
   b) Consider the following set of processes along with their burst time(in ms), arrival time(in ms). Determine the average waiting time and average turn around time using SJF(preemptive), RR (Quantum = 2) and HRRN. 8

Process	Arrival Time	Service Time (Burst Time)
P1	0	3
P2	3	6
P3	5	4
P4	6	5

3. a) Consider the traffic deadlock depicted in Figure below: 7
   i. Show that the four necessary conditions for deadlock indeed hold in this example.
   ii. State a simple rule for avoiding deadlocks in this system.



b) Consider the following Snapshot of a system: 7

Processes	Allocation ABC	Max ABC	Available ABC
P0	001	001	152
P1	100	175	
P2	135	235	
P3	063	065	
P4	001	065	

Answer the following questions using the Banker's algorithm:
   i. What is the content of the matrix need?
   ii. Is the system in a safe state? Also find the safe sequence.
   iii. If the request from process P1 arrives for (1, 0, 2), can the request be granted immediately?

4. a) Describe the actions taken by a kernel to context-switch between processes. Provide two programming examples in which multithreading does not provide better performance than a single-threaded solution? 7
   b) How many page faults occur for the following reference strings for 4 page frames: a, b, c, d, c, a, d, b, e, b, a, b, c, d Using LRU, FIFO and Optimal page replacement algorithm. Which algorithm performs better? 8
5. a) Given the following queue (in order): 95, 180, 34, 119, 11, 123, 62, 64 with the read-write head initially at track 50 and the trail track being at 199. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms? 7
   i. FCFS
   ii. SSTF
   iii. SCAN(initially moving upward)
   iv. CSCAN(initially moving downward
   b) What are the advantages and disadvantages of supporting memory mapped I/O to device control registers? 8
6. a) What is swap-space and how swap-space management is done? Explain with example. 7
   b) What are different file operations? Explain contiguous and linked list allocation in file systems. 8
7. Write short notes on: **(Any two)** 2×5
   a) DMA
   b) File Protection
   c) Stable storage implementation

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2018
Programme: BE Full Marks: 100
Course: Applied Operating System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is an operating system? Describe time sharing, real-time and distributed operating systems in brief. 8
   b) What is CPU bound and I/O bound processes? Explain blocking and non-blocking message passing and shared memory for Interposes communication. 7
2. a) Why is process synchronization needed? What problems may occurred if they are not synchronized? Describe Peterson's algorithm in detail 8
   b) What is deadlock? How do you determine the state of system is safe or unsafe using banker's algorithm for multiple resources type? 7
3. a) Assume you have the following jobs to execute with one processor, with the jobs arriving in the order listed here: 8

i	$T(P_i)$
0	80
1	20
2	10
3	20
4	50

Suppose a system uses SJN scheduling.
   i. Create a Gantt chart illustrating the execution of these processes?
   ii. What is the turnaround time for process p4?
   iii. What is the average wait time for the processes?

   b) What are the internal and external memory fragmentations? How are they resolved in paging? Explain in detail. 7
4. a) What is internal and external fragmentation? Consider the following page reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1. 8

How many page faults would occur for the following replacement algorithms, assuming three and four frames? In each case which algorithm perform better?
   i. LRU page replacement
   ii. FIFO page replacement
   iii. Optimal page replacement

   b) What are the different I/O techniques? Describe each in brief. 7
5. a) A disk has 8 sectors per track and spins at 600 rpm. It takes the controller 10 ms from the end of one I/O operation before it can issue a subsequent one. How long does it take to read all 8 sectors using the following interleaving systems? 7
   i. No interleaving
   ii. Single interleaving
   iii. Double interleaving

   b) Explain the I/O in UNIX system. 8
6. a) What are the different files access methods? Fxplain in detail. 7
   b) What is file system interface? Explain different file allocation methods. 8
7. Write short notes on: (Any two) 2×5
   a) System Call
   b) User and kernel threads
   c) RAID

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2019
Programme: BE Full Marks: 100
Course: Applied Operating System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define time sharing, parallel and real-time operating systems? How they are different from one another? Explain. 8
   b) Differentiate between process and thread. What are the benefits of threads? Explain different process states and possible transitions using a diagram. 7
2. a) Consider following set of processes along with their burst time, arrival time and priorities. Calculate average waiting time and average turnaround time using following scheduling algorithms. 8
   i. FCFS
   ii. SJF
   iii. Priority (Preemptive)
   iv. HRRN

Process	Arrival Time	Burst Time	Priority
A	0	3	5
B	2	6	4
C	4	4	1
D	6	5	3
E	8	2	2
F	3	4	1

   b) What is mutual exclusion? Show how mutual exclusion can be achieved using Peterson's Solution. 7
3. a) Define critical section problem. What is busy waiting? Explain semaphore and its use in critical-section problem with example. 7

   b) Suppose we have two resources, A, and B. A has 6 instances and B has 3 instances. Can the system execute the following processes without deadlock occurring? 8

Processes	Allocated A B	Maximum Need A B
P0	1 1	2 2
P1	1 0	4 2
P2	1 0	3 2
P3	0 1	1 1
P4	2 1	6 3

4. a) Define swapping. Explain contiguous and non-contiguous memory allocation scheme with their advantages and disadvantages. 8
   b) Define page fault. How many page faults occur for the following reference strings for 3 page frames: 3, 4, 5, 6, 5, 3, 6, 4, 7, 4, 3, 4, 5, 6, Using Second Chance, LRU, and FIFO replacement algorithm. 7
5. a) What are different ways to input/output? Explain Interrupt-Driven input/output with diagram. 7
   b) Given the following request queue (in order): 85, 170, 24, 109, 11, 123, 60, 62 with the head initially at track 50 and the trail track being at 184. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms? 8
   i. FCFS
   ii. SSTF
   iii. SCAN(initially moving inward)
   iv. C-SCAN(initially moving outward)
6. a) List out some operation on file. Explain different file access method with their advantages and disadvantages. 7
   b) What is file? Explain different free space management strategy in file systems. 8
7. Write short notes on: (Any two) 2×5
   a) OS services and system calls
   b) Demand paging and Thrashing
   c) Bad Sectors handling in OS.

POKHARA UNIVERSITY

Level: Bachelor   Semester: Spring   Year : 2019
Programme: BE   Full Marks: 100
Course: Applied Operating System   Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What are system calls? Explain different categories of system calls with example? 8

   b) What is process control block? How can a process and thread be differentiated? Elaborate process life cycle by providing its stage and life cycle. 7

2. a) What are semaphores? Explain solution to producer-consumer problem using semaphores. 7

   b) Consider a multilevel feedback queue scheduling (MLFBQ) with three queues q1, q2, and q3. q1 and q2 use round-robin algorithm with time quantum (TQ) = 5, and 4 respectively, q3 use first-come first-service algorithm. Find the average waiting time (A.W.T) and average turnaround time (A.T.A.T) for executing the following process? 8

Processes	P1	P2	P3	P4
Burst Time	8	22	4	12

3. a) Describe using bankers algorithm if the system is in safe state. 8

Process	Allocation	Max	Available
	A B C	A B C	A B C
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

   b) Consider following set of processes along with their burst time, arrival time and priorities. Calculate average waiting time and average turnaround time using following scheduling algorithms. 7

      i. FCFS

      ii. SJF
      iii. Priority (Preemptive)
      iv. HRRN

Process	Arrival Time	Burst Time	Priority
A	0	3	5
B	2	6	4
C	4	4	1
D	6	5	3
E	8	2	2

4. a) How many page faults occur for the following reference strings for 3 page frames: 8
   2, 3, 4, 5, 4, 2, 5, 3, 6, 3, 2, 3, 4, 5, Using MFU, LRU, and Optimal page replacement algorithm.

   b) Differentiate between paging and segmentation along with their figures. 7

5. a) Disk request come to the disk driver for cylinder 16, 18, 12, 6, 25, 38, 7 and 36 in that order. A seek take 2 micro sec per cylinder move. How much seek time is needed for 7
      i. FCFS
      ii. Closest Cylinder Next
      iii. C-Scan (Initially moving upward)
      iv. Scan (Initially moving downward)

   In all cases, the arm initially at cylinder 18. Also describe which one is best algorithm and why?

   b) What are different ways to input/output? Explain Interrupt-Driven i/o with diagram. 8

6. a) What are the different operations performed in files? Describe the ways by which directories can be accessed . 8

   b) How data is maintained and managed using RAID technology? Is there any chance of losing data using it? Describe each of available model in details. 7

7. Write short notes on: **(Any two)** 2×5
   a) Batch Systems
   b) Protection on files
   c) DMA

POKHARA UNIVERSITY

Level: Bachelor	Semester: Fall	Year : 2020
Programme: BE		Full Marks: 100
Course: Applied Operating System		Pass Marks: 45
		Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is an Operating System and how can it be used as a user interface? Explain with suitable diagram. Define multiprogramming, spooling in operating system. 8

   b) What is the critical section? What are the minimum requirements that should be satisfied by a solution to critical section problem? 7

2. a) What are semaphores? Explain solution to producer-consumer problem using semaphores. 7

   b) From the following set of informations, Compare the average waiting time and average turn-around time using FCFS, SJF(preemptive and non-preemptive), RR(quantum=2). 8

Process	Arrival Time	Burst Time
P1	0	10
P2	1	6
P3	2	12
P4	3	15

3. a) Define IPC. What are different methods used for logical implementations of message passing systems. 7

   b) Explain the algorithm of Resource Allocation Graph. Consider a system with three processes (P0-P2) and four allocable resources (A, B, C, D). The total four resources types in the amount as E= (4,2,3,1). The current allocation matrix and request matrix are as follows. 8

   Using Banker's algorithm find:

   i) What will be the context of need matrix?

   ii) Is the system in safe state? If yes, then what is the safe state sequence?

Current Allocation Matrix					Allocation Request Matrix				
Process	A	B	C	D	Process	A	B	C	D
P0	0	0	1	0	P0	0	1	2	1
P1	2	0	0	1	P1	0	0	1	0
P2	0	1	2	0	P2	1	0	2	0

4. a) Define page fault. How many page fault occur for the following reference string for four page frame using Optimal page replacement and LRU algorithms? reference string : 1,3,1,5,7,7,3,7,4,9,8,1,6,3,4,2,5,8 7

   b) Explain about Fixed partition and Variable partition. Consider a swapping system in which memory space is as: 8
   400, 700, 1200,250,300 bytes.
   File sizes are as: 900, 25, 600, 200, 300 bytes
   Find the total fragmentation using
   (i) First Fit (ii) Next fit (iii) Best fit (iv) Worst fit

5. a) Suppose the following request queue (in order): 74, 168, 23, 109, 10, 122, 59, 62 with the head initially at track 48 and the trail track being at 174. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms? 8

      i. FCFS
      ii. SSTF
      iii. SCAN (initially moving downward)
      iv. C-SCAN (initially moving upward

   b) Differentiate between Paging and Segmentation. Explain inverted page table. 7

6. a) What are the three different ways to do input- output? Explain all. 8

   b) What are the different methods used for accessing a file? Compare and contrast all the methods and explain which method is best in term of fast execution of the file. 7

7. Write short notes on: **(Any two)** 2×5

   a) RAID
   b) Direct Memory Access
   c) PCB
   d) Interrupt handling

POKHARA UNIVERSITY

Level: Bachelor | Semester – Spring | Year: 2020

Programme: BE | Full Marks : 70

Course: Applied Operating System | Time: 2 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.

Attempt all the questions.

Group –A: (5×10=50)

1. System calls are important for Operating System to perform. Justify your views. Explain different categories of system calls with example.

OR

Differentiate between interrupt driven i/o and Programmed i/o and DMA(Direct Memory Access).

2. Is semaphores can be used for achieving mutual exclusion? How? Explain solution to producer-consumer problem using semaphores.
3. A system has three types of resources R1 R2 R3 and their number of units are 3, 2, 2 respectively. Four processes P1 P2 P3 p4 are currently competing for these resources in following number.
   1. P1 is holding one unit of R1 and is requesting for one unit of R2.
   2. P2 is holding two units of R2 and is requesting for one unit each of R1 and R3.
   3. P3 is holding one unit of R1 and is requesting for one unit of R2.
   4. P4 is holding two units of R3 and requesting for one unit of R1.

   Determine which if any of the processes are deadlock in this state.
4. Is page fault is good for OS? Justify your views. How many page faults occur for the following reference strings for 3 page frames:

   8, 1, 9, 5, 4, 8, 5, 3, 6, 3, 8, 1, 9, 5, Using LFU, Second Chance, and Optimal page replacement algorithm.
5. Suppose the following request queue(in order): 73, 173, 24, 109, 11, 121, 59, 63 with the head initially at track 35 and the trail track being at 179. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms?
   - i. FCFS
   - ii. SSTF
   - iii. SCAN(initially moving downward)
   - iv. C-SCAN(initially moving upward)
   - v. LOOK

Group –B (1×20=20)

6. Consider the following set of processes along with their burst time(in ms), arrival time(in ms). Determine the average waiting-time and average turn-around time using SJF(preemptive), RR (Quantum = 2) ,HRRN and FCFS. Which algorithm is best.

Process	Arrival Time	Service Time (Burst Time)
P1	0	4
P2	4	7
P3	6	5
P4	7	6

What are points to be consider in file system design? Differentiate between linked list allocation, index allocation and contiguous allocation in file system.

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2021
Programme: BE Full Marks: 100
Course: Applied Operating System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define operating system. How operating system creates abstraction? Explain with reference to operating system as extended machine. 7

   b) What is process and process state? Do you think a process can exist without any state? Justify your answer with the help of a process state diagram and PCB. 8

2. a) Why semaphores are used? Give solution to reader-writer problem using semaphores. 7

   b) Suppose 5 batch jobs A, B, C, D and E arrived at the service center at time 0. They have burst time 20, 22, 8, 12 and 14 respectively. Their priorities are 4, 2, 1, 3 and 5 with 5 being the highest priority. For each of the scheduling algorithm determine average waiting time (WT) and average turn-around time (TAT) using: 8
      i. SJF
      ii. RR (Quantum size 10)
      iii. Priority (preemptive)

3. a) Consider following snapshot of a system 8

Processes	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	0	1	0	0	1	1	5	2
P1	1	0	0	1	7	5			
P2	1	3	5	2	3	5			
P3	0	6	3	0	6	5			
P4	0	0	1	0	6	5			

Answer the following questions using Banker's algorithm: 7
   i) What is the content of Need Matrix ?
   ii) Is the system in a safe state? Also find the safe sequence.

   b) Explain different multithreading operating system designs with advantages and disadvantages.

4. a) What is mutual exclusion? Show how mutual exclusion can be achieved using Dekkers Algorithm? 7

   b) How many page faults occur for following reference strings for 4 page frames? 8
      5, 0, 1, 2, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 1, 2, 0, 3
      Using FIFO, LRU and Optimal page replacement algorithm

5. a) Disk request come to the disk driver for cylinder 16, 18, 12, 6, 25, 38, 7 and 36 in that order. A seek take 5 micro sec per cylinder move. How much seek time is needed for 8
      i. FCFS
      ii. SSTF
      iii. C-SCAN (upward)
      iv. C-LOOK (downward)
   In all case, the arm is initially at cylinder 18.

   b) What is difference between Paging and Segmentation? What are different page-table structures? Explain any one in detail. 7

   a) Describe Interrupt. How Operating system handles the interrupt? Explain with the help of a block diagram 8

   b) What is file descriptor? Discuss about different file allocation methods. 7

Write short notes on: (Any two) 2×5
   a) Stable storage Implementation
   b) File Protection
   c) RAID

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2021
Programme: BE | Full Marks: 100
Course: Applied Operating System | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is system call? Define major services of operating system. 8
   b) What is CPU bound and I/O bound process? Why inter process communication is required? Explain. 7
2. a) Explain the algorithm of Resource Allocation Graph. Consider a system with three processes (P0-P2) and four allocable resources (A, B, C, D). The total four resources types in the amount as E= (4, 2, 3, 1). The current allocation matrix and request matrix are as follows. 8
   Using Banker's algorithm find:
   i) What will be the context of need matrix?
   ii) Is the system in safe state? If yes, then what is the safe state sequence?

Current Allocation Matrix					Allocation Request Matrix				
Process	A	B	C	D	Process	A	B	C	D
P0	0	0	1	0	P0	0	1	2	1
P1	2	0	0	1	P1	0	0	1	0
P2	0	1	2	0	P2	1	0	2	0

   b) For what purpose semaphores are used? Give solution to producer-consumer problem using semaphores. 7
3. a) How process differs from thread? Explain and differentiate between user and kernel thread. Draw figures to illustrate. 8
   b) What mutual exclusion, race condition and critical condition? Can Peterson's algorithm is guaranteed to solve critical condition problem. Justify your answer. 7

4. a) Define page fault. How many page fault occur for the following reference string for four page frame using Optimal page replacement and LRU algorithms? 8
   reference string : 1,3,1,5,7,7,3,7,4,9,8,1,6,3,4,2,5,8
   b) Explain about Fixed partition and Variable partition. Consider a swapping system in which memory space is as: 7
   400, 700, 1200,250,300 bytes.
   File sizes are as: 900, 25, 600, 200, 300 bytes
   Find the total fragmentation using
   i) First Fit
   ii) Next fit
   iii) Best fit
   iv) Worst fit
5. a) Consider the following request queue (in order): 80, 178, 24, 110, 11, 123, 61, 68 with the head initially at track 45 and the trail track being at 189. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests, for each of the following disk-scheduling algorithms? 8
   i. FCFS
   ii. SSTF
   iii. SCAN (initially moving outward)
   iv. C-SCAN (initially moving inward)
   b) What is fragmentation? Explain segmentation with paging with an example. 7
6. a) What are the three different ways to do input- output? Explain all. 7
   b) What are different file operations? Discuss different file access methods. 8
7. Write short notes on: (Any two) 2×5
   a) PCB
   b) Interrupt Handling
   c) DMA
   d) File security in Linux

POKHARA UNIVERSITY

Level: Bachelor — Semester: Fall — Year : 2022

Programme: BE — Full Marks: 100

Course: Applied Operating System — Pass Marks: 45

Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Describe the two basic purposes of Operating system. Differentiate time sharing system from real time system with examples. 7

   b) What is race condition and critical section? Give hardware solution to achieve mutual exclusion. 8

2. a) Producer Consumer problem can be solved with the help of various techniques. Suggest any one of the techniques to solve producer consumer with detailed explanation. 7

   b) Given the following information, draw GANTT charts and find the average waiting time and average turn-around time using FCFS, SRTN, RR (Quantum=3), HRRN. 8

Process	Arrival Time	Burst Time
A	0	9
B	3	3
C	4	4
D	6	10

3. a) What are the conditions of deadlock ? Use Banker algorithm to find whether the system is in deadlock or not? 8

Process	Allocation A B C	Maximum A B C	Available A. B. C
P1	0 1 0	7 5 3	3 2 3
P2	2 0 0	3 2 2	
P3	3 0 2	9 0 2	
P4	2 1 1	2 2 2	
P5	0 0 2	4 3 3	

   b) When multiple processes need to co-operate, there is a choice between shared memory and message passing. Compare and contrast these techniques. What is the role of the operating system in each? 7

4. a) What is semaphore? State and give solution to dining philosopher problem. 7

   b) Given references to the following pages by a program 0,4, 1, 4, 2, 4, 3, 4, 2, 4, 0, 4, 1, 4, 2, 4, 3, 4,. How many page faults will occur if the program has four page frames for each of the following algorithms? 8

      a. FIFO. b. Optimal. c. LRU d. Second chance

5. a) Discuss in detail the use of TLB in the process of paging. Support your answer with the help of a diagram. 7

   b) Consider the following request queue (in order): 72, 171, 26, 109, 10, 125, 62, 67 with the head currently at track 50 and the previous head track was at 45. Starting from the current head position, what is the total head movement to satisfy all the pending requests, for each of the following disk-scheduling algorithms? Assume disk size 180 cylinders. 8

      i. FCFS
      ii. SSTF
      iii. SCAN
      iv. LOOK

6. a) Explain Programmed Driven I/O and Interrupt Driven I/O with Suitable diagrams. 7

   b) What are different file attributes? Explain file system allocation. 8

7. Write short notes on: (Any two) 2×5

   a) Sector slipping v and Sector Sparing.

   b) RAID

   c) Directory Structures

POKHARA UNIVERSITY

Level: Bachelor Semester: Spring Year : 2023
Programme: BE Full Marks: 100
Course: Applied Operating System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

a) Define time sharing, parallel and real-time operating systems? How they are different from one another? Explain. 8

b) Describe the fields in a process control block (PCB) with diagram. What is switching overhead? 7

a) Define Semaphores. Explain solution to the Dining Philosophers Problem using semaphores. 7

b) Given the following information's, draw GANTT charts and find the average waiting time and average turn-around time using STRF, RR (quantum=4), HRRN. 8

Process	Arrival Time	Burst Time
A	0	10
B	3	4
C	6	6
D	7	9

a) How many page faults occur for the following reference strings for 3-page frames using Optimal, LRU, and Second Chance.? 8
4, 6, 7, 8, 7, 4, 8, 6, 9, 6, 4, 6, 7, 8. Also calculate fault ratio.

b) What do you mean by disk management? Compare sector Slipping and Sector Sparing in bad block recovery during disk management. 7

a) What is deadlock? Consider the following snapshots of the system having 10 instances of resource A, 5 instances of resource B and 7 instances of resource C. 8

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

Answer the following questions using the Banker's Algorithm:

i. Is the system in a safe state? Also find the safe sequence.

ii. If the request from process P1 arrives for (1, 0, 2) can the request be granted immediately?

b) What is Memory Fragmentation? What are its types? How can the problem of fragmentations be solved? 7

5. a) Suppose a disk drive has 300 tracks, numbered 0 to 299. The current position of the R/W head is at track 160 and the previous request was at track 145. The sequence of pending requests is 43,172, 150, 48, 85, 270, 290, 130. Starting from current position what is the total number of track movements (distance) for the following disk scheduling algorithms. 8

i. SSTF
ii. SCAN
iii. C-LOOK
iv. FCFS

b) Explain interrupt driven I/O with diagram. How is it different from polled I/O? 7

6. a) What are different file attributes? Discuss different file allocation methods. 8

b) What is File? Compare the characteristics of various file access methods. 7

7. Write short notes on: (Any two) 2×5

a) RAID
b) System Calls
c) Deadlock vs. Starvation

POKHARA UNIVERSITY

Level: Bachelor Semester: Fall Year : 2023
Programme: BE Full Marks: 100
Course: Applied Operating System Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What are main purposes of Operating Systems? Can OS be called as a resource manager and an extended machine? Justify. 8
   b) What is a process? Explain the different states of process with a neat diagram. 7
2. a) Define semaphores. How Dining Philosopher problem can be solved using semaphores? Explain with necessary pseudo code. 7
   b) Consider following set of processes along with their burst time, arrival time and priorities. Calculate average waiting time and average turnaround time using following scheduling. Also describe which one is the best algorithm and why? 8
      i. SRTF
      ii. Priority
      iii. RR (Quantum size = 3ms)

Process	Burst Time	Arrival Time	Priority
P1	3	0	2
P2	14	1	1
P3	9	2	3
P4	17	3	4

3. a) Consider following snapshot of a system 8

Processes	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
A	0	1	1	1	2	3	0	1	0
B	1	1	0	2	2	0			
C	0	0	1	0	1	1			
D	1	2	1	3	5	3			
E	1	0	1	1	1	2			

Answer the following questions using Banker's algorithm:
   i. What is the content of the Need Matrix?
   ii. Is the system in a safe state? Also find the safe sequence.

   b) Define thread. Why is it called a light weight process? Compare and contrast between process and thread. 7
4. a) Explain race condition and critical condition. Can Peterson's algorithm be guaranteed to solve critical condition problem. Justify your answer. 7
   b) What is page fault? Given below is the reference made to the following pages by a program: 1,3,1,5,7,7,3,7,4,9,8,1,6,3,4,2,5,8. Show the successive page residing in the four frames using replacement policy below: 8
      i. Second Chance
      ii. LRU
      iii. Optimal
5. a) The disk track requests are: 111, 123, 250, 298, 120, 13, 288 and 224. There are 300 cylinders numbered from 0 to 299. Assume that the last request is at track 130 and the head is moving towards track 0. Find out the total seek time for each of the disk scheduling algorithms below: 8
      i. SSTF
      ii. C-SCAN
      iii. LOOK
   b) Difference between Fixed partition and Variable partition. Consider a swapping system in which memory space is as: 400, 750, 1200,250,150 bytes. 7
      File size are as: 200, 75, 600, 900,300
      Find the total fragmentation using
      i. First Fit
      ii. Next fit
      iii. Best fit
      iv. Worst fit
6. a) What are the problems of programmed and interrupt driven I/O techniques? How does DMA solve these problems? Explain in detail. 8
   b) Explain all methods used for file allocation. Compare and contrast all the methods with appropriate diagrams. 7
7. Write short notes on: (Any two) 2×5
   a) Free space management
   b) Scan vs CScan
   c) Stable Storage Implementation

POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2024
Programme: BE | Full Marks: 100
Course: Applied Operating System | Pass Marks: 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Define OS as a resource manager and an extended machine. Differentiate between batch system and time-sharing system. 8
   b) Explain the lifecycle of a process with the help of a process state diagram. 7
2. a) Define Semaphores. Explain solution to the Producer Consumer Problem using semaphores. 7
   b) Given the following information's, draw GANTT charts and find the average waiting time and average turn-around time using FCFS, RR (quantum=3), SRTF, HRRN. 8

Process	Arrival Time	Burst Time
P1	0	9
P2	3	4
P3	4	5
P4	6	10

3. a) What is deadlock? Find the safe sequence if any, using the Bankers Algorithm for the following information. 8

Process	Allocation	Max	Available
	A B C D	A B C D	A B C D
P0	6 0 1 2	4 0 0 1	3 2 1 1
P1	2 7 5 0	1 1 0 0	
P2	2 3 5 6	1 2 5 4	
P3	1 6 5 3	0 6 3 3	
P4	1 6 5 6	0 2 1 2	

   b) Explain race condition and critical condition. Can Peterson's algorithm is guaranteed to solve critical condition problem. Justify your answer. 7

4. a) Difference between Fixed partition and Variable partition. Given the following free memory blocks: 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB. If a new process requires 212 KB, 350 KB, 525 KB, 85 KB use the first-fit, best-fit, next fit and worst-fit memory allocation strategies to determine which block will be allocated to the process. Also, calculate the resulting total fragmentation for each strategy. 7
   b) Define page fault. How many page faults occur for the following reference strings for 3-page frames? 5, 6, 7, 8, 7, 5, 8, 6, 9, 6, 5, 6, 7, 8. Using Optimal, Second Chance, and LRU. Also calculate fault ratio. 8
5. a) Consider a disk with 200 tracks (numbered from 0 to 199). The disk head is initially at track 50. The requested tracks, in the order they arrive, are: 95, 180, 34, 119, 11, 123, 62. Calculate the total head movement using: 8
      i. C-SCAN
      ii. FCFS
      iii. SSTF
      iv. LOOk
   b) What are different ways to input output? Explain DMA with suitable diagram. 7
6. a) What is File? Explain i-node file system implementation. 7
   b) What are different file attributes? Explain Contiguous and Linked List methods of file allocation. 8
7. Write short notes on: (Any two) 2×5
   a) Swap- space Management
   b) Context Switching
   c) RAID

POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2024
Programme: BE | Full Marks : 100
Course: Applied Operating System | Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is OS and what are its main responsibilities? Explain OS as an user extended machine with suitable diagram. 8

   b) Explain the lifecycle of a process with the help of a process state diagram. 7

2. a) Consider following set of processes along with their burst time, arrival time and priorities. Calculate average waiting time and average turnaround time using following scheduling. Also describe which one is the best algorithm and why? 8

   i. SJF (Preemptive)
   ii. Priority
   iii. RR (Quantum size = 3ms)

Process	Burst Time	Arrival Time	Priority
P1	7	0	2
P2	7	0	1
P3	6	2	3
P4	5	5	4

   b) Define Semaphores. Explain producer consumer problem and solution using semaphores. 7

3. a) What is deadlock? Find the safe sequence if any, using the Bankers Algorithm for the following information. 8

Process	Allocation	Max	Available
	A B C D	A B C D	A B C D
P0	6 0 1 2	4 0 0 1	3 2 1 1
P1	2 7 5 0	1 1 0 0	
P2	2 3 5 6	1 2 5 4	
P3	1 6 5 3	0 6 3 3	
P4	1 6 5 6	0 2 1 2	

   b) Explain page fault. Given below is the reference made to the following pages by a program: 2, 3, 2, 1, 5, 2, 4, 5, 3, 2, 5, 2, 1, 4, 6, 2, 3. Show the successive page residing in the three frames using replacement policy below: 7

   i. FIFO
   ii. LRU
   iii. Optimal

4. a) Differentiate between internal and external fragmentation? Explain TLB in detail. 7

   b) Difference between Fixed partition and Variable partition. Given the following free memory blocks: 100 KB, 500 KB, 200 KB, 300 KB, and 600 KB. If a new process requires 212 KB, 350 KB, 525 KB, 85 KB use the first-fit, best-fit, next fit and worst-fit memory allocation strategies to determine which block will be allocated to the process. Also, calculate the resulting total fragmentation for each strategy. 8

5. a) Explain interrupt driven I/O with diagram. How is it different from polled I/O? 7

   b) Consider the following request queue (in order): 72, 178, 34, 109, 10, 135, 58, 89 with the head currently at track 34. Starting from the current head position, what is the total head movement to satisfy all the pending requests, for each of the following disk-scheduling algorithms? Assume disk size 180 cylinders. 8

   i. C-SCAN
   ii. SSTF
   iii. LOOK
   iv. FCFS

6. a) Explain Kernel I/O subsystem. 8

   b) What are different file attributes? Discuss about different free space management strategy in file systems. 7

7. Write short notes on: **(Any two)** 2×5

   a) System Calls
   b) File Access Methods
   c) Resource Allocation Graph

POKHARA UNIVERSITY

Level: Bachelor   Semester: Spring   Year : 2025
Programme: BE   Full Marks : 100
Course: Applied Operating Systems   Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1\. a) How time-sharing systems and parallel systems differ in terms of operation? Explain the different structure of Operating System. 7

b) What is Mutual Exclusion? How do you protect the critical section of a process from Race Condition? 8

2\. a) Define Semaphores. Explain solution to the Dining Philosophers Problem using semaphores. 8

b) Given the following information's, draw GANTT charts and find the average waiting time and average turn-around time using STRF, RR (quantum=2), HRRN, FCFS. 7

Process	Arrival Time	Burst Time
A	0	8
B	3	5
C	5	7
D	7	11

3\. a) What is the difference between deadlock and starvation? Consider a system has octa core microprocessor (M), 20 storage devices(S), 10 IO devices(I) and 15 peripheral devices(P). The snapshot of the system is given below. 8

Process	Allocated	Requested
	S I P M	S I P M
P1	5 2 4 1	3 3 3 3
P2	3 3 6 2	3 6 9 3
P3	1 1 2 1	1 3 4 1
P4	3 1 2 1	2 1 2 1
P5	4 3 1 1	4 3 2 2

i. What is the content of need matrix.

ii. Use Banker's algorithm to determine whether the system is at safe state or not.

b) What are the problems in contiguous memory? Compare fixed partition with variable partition with suitable example. 7

4\. a) How many page faults occur for the following reference strings for 3-page frames? 8, 6, 7, 2, 7, 8, 2, 6, 9, 6, 8, 6, 7, 2, 6. Using LFU, Second Chance and LRU. Also calculate fault ratio. 8

b) Explain interrupt driven I/O with diagram. How is it different from programmed I/O? 7

5\. a) What is seek time? If disk request come to the disk driver for cylinder 16, 18, 12, 6, 25, 38, 7 and 36 in that order. A seek take 5 micro sec per cylinder move. How much seek time is needed for 8

i. SSTF

ii. C-SCAN (upward)

iii. C-LOOK

In all cases, the arm is initially at cylinder 18.

b) What is the role of device drivers? Explain how I/O requests are handled using interrupts. How DMA is more effective solution for IO handling? 7

6\. a) Explain directory implementation techniques and their impact on file system performance. 8

b) What is file and folder? Explain Contiguous and Linked List methods of file allocation. 7

7\. Write short notes on: **(Any two)** 2×5

a) I/O in Unix

b) File Attributes

c) Swap Space Management

POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2025

Programme: BE | Full Marks : 100

Course: Applied Operating System | Pass Marks : 45

Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define OS as a resource manager and an extended machine. Differentiate between batch system and time-sharing system. 7

   b) Explain the lifecycle of a process with the help of a process state diagram. 8

2. a) Producer Consumer problem can be solved with the help of various techniques. Suggest any one of the techniques to solve producer consumer with detailed explanation. 8

   b) Given the following information, draw GANTT charts and find the average waiting time and average turn-around time using FCFS, SRTN, RR (Quantum=3), HRRN. 7

Process	Arrival Time	Burst Time
A	0	9
B	3	3
C	4	4
D	6	10

3. a) Consider following snapshot of a system 7

Processes	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	0	1	0	0	1	1	5	2
P1	1	0	0	1	7	5			
P2	1	3	5	2	3	5			
P3	0	6	3	0	6	5			
P4	0	0	1	0	6	5			

Answer the following questions using Banker's algorithm:

i. What is the content of Need Matrix?

ii. Is the system in a safe state? Also find the safe sequence.

   b) What is dispatcher and dispatch latency? Differentiate short-term scheduler, medium term scheduler and long-term scheduler. 8

4. a) How many page faults occur for the following reference strings for 3-page frames using Optimal, LRU, and Second Chance? 8

   4, 6, 7, 8, 7, 4, 8, 6, 9, 6, 4, 6, 7, 8. Also calculate fault ratio.

   b) What is difference between Paging and Segmentation? What are different page-table structures? Explain any one in detail. 7

5. a) Consider the following request queue (in order): 72, 171, 26, 109, 10, 125, 62, 67 with the head currently at track 50 and the previous head track was at 45. Starting from the current head position, what is the total head movement to satisfy all the pending requests, for each of the following disk-scheduling algorithms? Assume disk size 180 cylinders. 8

   i. FCFS

   ii. SSTF

   iii. SCAN

   iv. LOOK

   b) Explain Programmed Driven I/O and Interrupt Driven I/O with Suitable diagrams. 7

6. a) What is bad sector? Describe the different bad block recovery mechanisms in disk management. 8

   b) What are different file attributes? Explain Contiguous and Linked List methods of file allocation. 7

7. Write short notes on: **(Any two)** 2×5

   a) System Calls

   b) Directory Structures

   c) RAID